

DAY 2 WEEK 1

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PROJECT: Battery SoC Estimation & Remaining Charging-Time Prediction Using Machine Learning

Day 2: Literature Review, Candidate Method List

1. Literature Review Purpose

The Day-2 review studies machine-learning approaches used for battery SoC estimation and remaining charging-time prediction. The goal is to understand existing modelling directions and use that understanding to select candidate methods for later experiments.

2. Literature Findings

Stacked Ensemble (RF + GBDT + GPR)

Combining Random Forest, Gradient Boosting Decision Trees and Gaussian Process Regression was identified as an approach that improved SoC estimation accuracy.

LightGBM with Feature Engineering

LightGBM with feature engineering was identified as effective for nonlinear battery and temperature effects.

Five-Algorithm SoC Comparison

Random Forest, GBM, Extra Trees, XGBoost and Decision Tree were compared, with ensemble approaches performing better in the reviewed work.

XGBoost + Random Forest Fusion

Combining XGBoost and Random Forest was identified as useful for improving SoC prediction for nonlinear battery behaviour.

Deep Learning Model Comparison

ANN and LSTM were identified as capable of high accuracy, while requiring more computation.

RNN / LSTM / GRU Methods

These recurrent models can capture time-dependent changes during battery charging.

EV Charging-Time Prediction

ELM, FFNN and SVR were identified as methods showing that machine learning can predict remaining charging time.

Hybrid Physics + Reinforcement Learning

Combining physics-based information with reinforcement learning was identified as an approach that can improve charging-time prediction, especially near full charge.

1. Comparison Table

Model	How it Works	Key Idea	Advantages	Limitations
Linear Regression	Finds a straight line that best fits the input and output data.	$y = mx + c$ (learns the best values of m and c)	Simple and easy to implement; Fast training; Good baseline	Cannot capture complex patterns; Assumes linear relationship
Decision Tree	Uses simple rules based on feature values to make predictions.	If-else rules (conditions)	Easy to understand; Handles non-linear patterns; Works well with mixed features	Can overfit (if too deep); Less stable (small data changes)
Random Forest	Combines many decision trees and takes the average of their predictions.	Multiple decision trees (ensemble)	More accurate than a single tree; Handles non-linear patterns; Less overfitting	Slightly slower than a single tree; Less interpretable

XGBoost	Builds trees one after another, correcting the errors of the previous trees using a boosting technique.	Sequential boosting (weaker learners + error correction)	High accuracy; Handles complex patterns; Works well with structured data	More complex to tune; Slower to train; Less interpretable
Model	How it Works	Key Idea	Advantages	Limitations
Artificial Neural Network (ANN)	Learns complex patterns using connected layers of neurons.	Input → Hidden → Output layers	Captures complex non-linear relationships; High accuracy; Good for large datasets	Needs more data; Slower training time; Less interpretable

2. Candidate ML Methods

- **Linear Regression** — Used as a simple baseline to predict battery SoC and charging time.
- **Decision Tree Regression** — Predicts the output using simple decision rules based on battery features.
- **Random Forest Regression** — Combines many decision trees to give more accurate and stable predictions.
- **XGBoost** — Uses boosted decision trees to capture complex battery patterns and improve prediction accuracy.
- **Artificial Neural Network (ANN)** — Learns complex relationships in the battery data and is used if traditional models are not sufficient.